Paper II – Urology (MCh Urology, Summer 2016)

Contents

1. II-1. Carcinoma bladder: aetiology, newer investigations and treatment (LAQ)
2. II-2. Urethral stricture: aetiology, presentation and management (LAQ)
3. II-3a. Transureteroureterostomy
4. II-3b. Acquired renal cystic disease
5. II-3c. Bladder diverticulum
6. II-3d. Botulinum neurotoxin
7. II-3e. Catheter-free trial (trial without catheter)
8. II-3f. Urethral caruncle

II-1. CA bladder, etiology, newer investigations and treatment.

Type: LAQ (25 marks)

Introduction

- Commonest urinary tract cancer; over 90% **UC (urothelial carcinoma)**. Men 3 times commoner; women present later. About 70% are **NMIBC (non-muscle-invasive bladder cancer)**; it recurs often.

Aetiology (risk factors)

Factor	Key facts
Smoking	30–40% of UC; risk 2–3 times. Aromatic amines form DNA adducts; **NAT2 (N-acetyltransferase 2) slow acetylators** at higher risk. Risk nears never-smokers >15 years after quitting
Occupation (5–10%)	Aromatic amines (dye, rubber, leather, painters, hairdressers); polycyclic hydrocarbons (aluminium, chimney sweeps)
Environment	**Arsenic** in water (West Bengal, Bangladesh, Taiwan)
Radiation	Pelvic radiotherapy 1.5–4 times; latency 15–30 years
Drugs	**Cyclophosphamide** (dose-dependent, above 20 g); pioglitazone (conflicting)
Chronic inflammation	*Schistosoma haematobium*, long-term catheter, stones → **SCC (squamous cell carcinoma)**; exstrophy → adenocarcinoma
Genetic	First-degree relatives 2 times; **Lynch syndrome** (MSH2); GSTM1 deletion

- **Also know:** obesity raises risk; diet has no proven role.

Pathology and staging

- **WHO (World Health Organization) 2004 grading**: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. Grade predicts progression.
- **Papillary pathway** (FGFR3; recurs, rarely progresses) versus **flat CIS (carcinoma in situ)** (TP53, RB; progresses).

TNM (tumour, node, metastasis), 2017	Extent
Ta / Tis	Papillary non-invasive / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle

TNM (tumour, node, metastasis), 2017	Extent
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	One pelvic node / several / common iliac

Clinical features and standard work-up

- **Painless gross haematuria (85%)**; irritative symptoms suggest CIS.
- **Cystoscopy and biopsy (gold standard)**; **CT (computed tomography) urography**; **urine cytology** (sensitivity 31–62%, specificity 94–100%, detects high grade; **Paris System** reporting).

Newer investigations

Test	Principle	Value
Blue-light cystoscopy (PDD, photodynamic diagnosis) with HAL (hexaminolevulinate)	Porphyrins in tumour fluoresce red under blue light	About one-third more CIS; recurrence benefit inconsistent
NBI (narrow-band imaging)	415 and 540 nm light absorbed by haemoglobin highlights vessels	More tumours; many false positives
Urine biomarkers	NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), **UroVysion FISH (fluorescence in situ hybridisation)** (chromosomes 3, 7, 17, 9p21), ImmunoCyt, Cxbladder	More sensitive, less specific than cytology; **none replaces cystoscopy**
mpMRI (multiparametric MRI) with VI-RADS (Vesical Imaging-Reporting and Data System)	Muscle-line integrity scored 1–5	**≥3 suspicious for muscle invasion**; about 80–90% accurate; before TURBT
En bloc TURBT (transurethral resection of bladder tumour)	Tumour removed in one piece with detrusor	Better staging and muscle sampling, fewer perforations; recurrence similar

- **Also know:** optical coherence tomography; urinary tumour DNA; PET-CT (positron emission tomography) for doubtful metastases.

Treatment

1. TURBT: complete, with muscle in the specimen. **Re-TURBT within 6 weeks** for T1, incomplete resection or no muscle. **Single chemotherapy instillation within 6 hours** (mitomycin C, epirubicin or gemcitabine) for low-risk tumours; never BCG (bacillus Calmette–Guérin) at this time.

2. NMIBC by risk (AUA (American Urological Association))

Risk	Definition	Treatment
Low	Solitary low-grade Ta ≤3 cm; PUNLMP	Single instillation; surveillance
Intermediate	Low-grade Ta recurrent, >3 cm or multifocal; low-grade T1; high-grade Ta ≤3 cm	Chemotherapy or BCG; maintenance 1 year
High	High-grade T1; large, multifocal or recurrent high-grade Ta; any CIS; variant histology; lymphovascular invasion	**BCG induction + 3-year maintenance**; early cystectomy if very high risk

- **BCG** induces a Th1 immune response; six weekly doses from 2–4 weeks after TURBT, then **SWOG (Southwest Oncology Group) maintenance** (recurrence-free survival 76.8 vs 35.7 months). Only agent that reduces progression.
- **BCG-unresponsive**: radical cystectomy; if unfit, **pembrolizumab** (KEYNOTE-057), **nadofaragene firadenovec**, gemcitabine–docetaxel, TAR-200.

3. MIBC (muscle-invasive bladder cancer), cT2–T4a N0M0

- **Cisplatin-based NAC (neoadjuvant chemotherapy)** (dose-dense MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) or gemcitabine–cisplatin) → **radical cystectomy + pelvic lymphadenectomy**. NAC adds 5–8% absolute survival (SWOG 8710: pT0 38% vs 15%). Cisplatin-ineligible: GFR (glomerular filtration rate) <60 mL/min, poor performance status, hearing loss, neuropathy, heart failure.
- **Diversion**: ileal conduit or orthotopic neobladder. Robotic equals open (**RAZOR**).
- **Trimodality therapy**: complete TURBT + chemoradiation (5-FU (fluorouracil) + mitomycin C, **BC2001**). For solitary T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- **Adjuvant nivolumab** for ≥pT3 or N+ (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months). Partial cystectomy for selected solitary dome tumours.

4. Metastatic: **enfortumab vedotin + pembrolizumab** (**EV-302**: nearly doubled survival versus platinum). Otherwise platinum then **avelumab maintenance**; **erdafitinib** for FGFR3 alterations.

Follow-up and recent advances

- Cystoscopy at 3 months, then by risk; upper-tract imaging in high risk.
- Molecular subtypes (basal tumours respond to cisplatin); **NIAGARA** (2024): perioperative durvalumab with NAC improved survival.

Diagram to draw: bladder wall layers with Ta, Tis, T1–T4 marked.

> **Key points to remember:**
>
> - Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
> - Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
> - High-risk NMIBC: BCG with 3-year maintenance.
> - MIBC: cisplatin NAC + cystectomy; trimodality in selected.

II-2. Describe etiology, clinical presentation and management of urethral strictures.

Type: LAQ (25 marks)

Definition

- A fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing mucosa. It is an epithelial scar with **spongiofibrosis**.
- Posterior narrowing is named differently: after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery, **stenosis or contracture**.
- Area falls with the square of diameter: halving calibre (30 to 15 Fr) loses three-quarters of the lumen, so symptoms appear late.

Anatomy relevant to surgery

- **Bulbar urethra lies eccentric (dorsal)** in the spongiosum; the penile urethra is central.
- **Dual blood supply** (bulbar arteries and retrograde flow from the glans) allows the bulb to be divided and mobilised.

Aetiology

Category	Examples
Idiopathic (commonest, with iatrogenic)	No cause found
Iatrogenic	Catheters (common in LMICs (low- and middle-income countries)), TURP (transurethral resection of prostate), bladder tumour resection, cystoscopy, hypospadias repair, radiotherapy
Traumatic	**Straddle injury**: bulb crushed against the pubis; pelvic fracture (PFUI)
Inflammatory	Gonorrhoea (now rare); **lichen sclerosus (LS)**
Congenital	Rarest; infants only

- **LS**: chronic skin disease starting at the glans and meatus. Urine forced into Littre glands causes inflammation and deep spongiofibrosis. It recurs in any skin used to repair it.
- **Jordan's spectrum**: mucosal fold → iris → full-thickness → deep spongiofibrosis → periurethral fibrosis → fistula or abscess.

Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- Retention after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

Evaluation (define 4 things: location, length, depth, density)

Test	Role
Uroflowmetry and PVR (post-void residual)	Low, plateau-shaped flow; screen and follow-up
Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)	Site and length; steep oblique views; done by the surgeon
Flexible urethroscopy	Calibre, appearance; paediatric scope sees beyond
Sonourethrography	Length and depth of spongiofibrosis

Test	Role
MRI (magnetic resonance imaging)	PFUI defect alignment; not routine

- **Urethral rest** (suprapubic catheter 6–8 weeks) shows the true length. Avoid blind dilation. In retention, place a **suprapubic catheter** rather than force a dilation.

Management

First agree the aim: **palliation or cure**. The old "reconstructive ladder" (repeat endoscopy before surgery) is abandoned, because repeated endoscopy makes later urethroplasty harder.

1. Dilation: soft or balloon dilation over a guidewire. It stretches, not tears, the scar. Curative only for epithelial strictures without spongiofibrosis.

2. DVIU (direct vision internal urethrotomy)

- Heals by **secondary intention**: a race between epithelialisation and wound contraction.
- Best for **bulbar strictures <1.5 cm**, without dense spongiofibrosis, at first attempt. A second urethrotomy adds almost nothing. Laser gives no advantage over cold knife.
- Complications: recurrence (commonest), bleeding, extravasation, ED (erectile dysfunction) from deep cuts.
- Adjuncts: **intermittent self-dilation**; mitomycin C injection. **Optilume** paclitaxel-coated balloon (**ROBUST III**: 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy).

3. Urethroplasty

Technique	Indication and key points
Excision and primary anastomosis (EPA)	**Gold standard**, about 90–95% success; short bulbar strictures (1–2 cm, up to 3–5 cm proximally). Excise all scar, spatulate widely, tension-free. **Non-transecting** and **vessel-sparing** variants preserve bulbar arteries
BMG (buccal mucosal graft) onlay	Longer bulbar strictures: **dorsal onlay (Barbagli)**, ventral onlay, **dorsal inlay (Asopa)**. Graft equals flap. Kulkarni one-sided repair for panurethral strictures
Augmented anastomosis	Excise the narrowest part, anastomose the floor, graft the roof
Penile skin flaps (Orandi, Quartey, circular)	Penile strictures; skin island carried on dartos
Staged repair (Johanson)	Stage 1 open and graft; tubularise after 6 months or more. For **LS**, failed hypospadias, infection
Perineal urethrostomy	Complex or panurethral disease, elderly or after failures

- **Onlay is better than tube** repairs; **avoid genital skin in LS** (use oral mucosa). Topical clobetasol for early LS meatal disease.
- Meatal stenosis: meatotomy; fossa navicularis: flap or graft.
- **PFUI**: suprapubic catheter, then delayed **perineal anastomotic urethroplasty** after 3 months or more. Progressive steps: bulbar mobilisation, intracrural (corporal) separation, inferior pubectomy, rerouting of the urethra.
- **Urethral stents** (UroLume) are withdrawn; AUA (American Urological Association): no stent if urethroplasty is possible.

Complications of urethroplasty

- Recurrence, ED (overall similar to circumcision; higher with long repairs), chordee, post-void dribble, graft-site pain, numbness.

Follow-up

- Uroflow, PVR and symptom score; RUG or cystoscopy if flow falls. Most EPA recurrences occur in the first 6 months; graft results decline with time.

Recent advances

- Tissue-engineered oral mucosa; robotic reconstruction of posterior stenosis.

Diagram to draw: RUG of a bulbar stricture, and cross-section of dorsal onlay versus ventral onlay.

> **Key points to remember:**
> - Idiopathic and iatrogenic are commonest; LS recurs in skin.
> - Assess location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
> - DVIU only for bulbar <1.5 cm at first attempt; do not repeat.
> - EPA is gold standard; BMG onlay for longer; staged repair for LS.

II-3a. Trans Uretero Ureteric anastomoses

Type: SAQ (10 marks)

Definition

- **TUU (transureteroureterostomy)**: the injured (donor) ureter is brought across the midline and joined **end-to-side** to the opposite (recipient) ureter. First clinical use in 1934.

Indications

- Lower ureteric loss that reimplantation, psoas hitch or Boari flap cannot bridge: small, scarred or irradiated bladder; long ureteric loss; pelvic phlegmon; associated rectal, vascular or extensive bladder injury.
- One ureter short while the other is reimplanted with a psoas hitch (hitch only one side).
- Undiversion and reconstruction in children (Hendren).
- Usually a **delayed** repair, not acute.

Contraindications

Absolute	Relative (disease that may involve both ureters)
Donor ureter too short to reach without tension	Stone disease, **urothelial carcinoma**
Diseased recipient ureter	Retroperitoneal fibrosis, pelvic radiation
	Chronic pyelonephritis, genitourinary tuberculosis

- Recipient reflux must be corrected at the same time, so do a preoperative VCUG (voiding cystourethrogram). Less favoured in children (long life, both kidneys at risk).

Technique (steps)

1. Midline transperitoneal incision; reflect the colon.
2. Mobilise the **donor ureter with its adventitia** (blood supply); divide it just above the diseased segment.
3. Expose about 5 cm of recipient ureter with minimal mobilisation.
4. Tunnel the donor ureter **behind the sigmoid mesentery, above the IMA (inferior mesenteric artery)** to avoid kinking.
5. **Anteromedial ureterotomy** of the recipient (at least 2 cm); **spatulated, tension-free end-to-side** anastomosis with fine absorbable interrupted sutures.
6. **Double-J stent** from donor kidney to bladder; drain. (Laparoscopic and robotic routes are feasible.)

Complications

- Urine leak (commonest), urinoma, stricture or obstruction, kinking under the IMA.
- **Main risk: one unilateral problem becomes bilateral obstruction.**
- Later endoscopic access is difficult; use the antegrade route.

Results

- Success 90–97% in adults; complications (mostly leak) in about one-quarter of complex cases.

Diagram to draw: donor ureter crossing above the IMA to an end-to-side anastomosis, with stent.

> **Key points to remember:**
>
> - End-to-side, spatulated, tension-free; cross above the IMA.
> - Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
> - Risk: unilateral injury becomes bilateral.

II-3b. Acquired renal cystic disease

Type: SAQ (10 marks)

Definition

- **ARCD (acquired renal cystic disease)**: bilateral cysts in kidneys of patients with **ESRD (end-stage renal disease)** not due to inherited cystic disease. Diagnosis: **≥3 cysts in each kidney**.
- Described in 1977; occurs with haemodialysis and peritoneal dialysis.

Epidemiology

Duration of renal failure	Prevalence
Before dialysis	About 10%
3 years of dialysis	44%
5 years	60%
≥10 years	>90%

- Male:female 3:1; independent of the cause of renal failure.

Aetiology

- **Uraemia, not dialysis**, is the cause; dialysis only gives cysts time to form.
- Tubular obstruction (fibrosis, oxalate crystals); uraemic toxins; growth factors (EGF (epidermal growth factor)); compensatory hyperplasia.
- Cysts may regress after transplant (data conflict); cancer risk persists.

Clinical features

- Mostly asymptomatic.
- **Bleeding into cysts (up to 50%)**: loin pain, haematuria, retroperitoneal haematoma; worsened by uraemic coagulopathy and heparin.
- Cyst infection, stones, polycythaemia (erythropoietin).
- **Neoplasia**: adenomas in 20–25%. **RCC (renal cell carcinoma)** risk about 3 times that without ARCD. **Acquired cystic disease-associated RCC** is a distinct subtype (oxalate crystals, good prognosis); papillary RCC is also common.

Evaluation

- **Ultrasound**: small echogenic kidneys, many small cysts.
- **CT (computed tomography) with and without contrast**: best for tumours. MRI (magnetic resonance imaging) for small lesions; gadolinium risks NSF (nephrogenic systemic fibrosis).

Feature	ARCD	ADPKD (autosomal dominant polycystic kidney disease)
Kidney size	Small (<300 g)	Large (>800 g)
Cysts	Small	Large
Extrarenal	None	Liver cysts, aneurysms

Management

- **Bleeding**: conservative; persistent → **embolisation or nephrectomy**; peritoneal dialysis avoids heparin.
- Painful cyst: aspiration; infected cyst: drainage, nephrectomy if it fails.

- **Tumour >3 cm: nephrectomy**. <3 cm: nephrectomy in fit patients, or yearly CT and excise if growing.
- **Screening** (debated): younger fit patients after 3 years of dialysis, and before transplant.
- After transplant, native-kidney RCC is more aggressive; >70% of transplant RCCs arise there.

Diagram to draw: small kidney with multiple cysts and a small solid tumour.

> **Key points to remember:**
> - ≥3 cysts per kidney in ESRD; >90% after 10 years.
> - Uraemia causes it; transplant may regress cysts, not risk.
> - RCC risk 3 times; image dialysis patients with haematuria.

II-3c. Bladder diverticulum.

Type: SAQ (10 marks)

Definition

- Herniation of bladder mucosa through the **muscularis propria**, forming a sac with a neck. With no functioning muscle it **cannot empty** (stasis).
- Male:female about 9:1.

Classification

Feature	Congenital (primary)	Acquired (secondary)
Age	Children (<10 years)	Men >60 years
Cause	Weak detrusor at the ureteric hiatus	**BOO (bladder outlet obstruction)**: BPH (benign prostatic hyperplasia) in about 70%, stricture; neurogenic bladder
Number, site	Solitary; **paraureteral (Hutch)**	Often multiple, anywhere
Bladder wall	Smooth	Trabeculated
Associations	VUR (vesicoureteral reflux); Ehlers–Danlos, Williams, Menkes syndromes	Stones, cancer

- **Also know:** iatrogenic (after cystotomy or reimplantation); cellule, saccule and diverticulum form a continuum.

Clinical features

- Usually incidental while investigating LUTS (lower urinary tract symptoms), haematuria or UTI (urinary tract infection).
- **Double voiding**, recurrent UTI, stones, retention, ureteric obstruction.
- **Malignancy** in 0.8–10% (urothelial 70–80%, squamous 20–25%). With no muscle layer, tumours invade the fat early (beyond lamina propria = **T3a**).

Investigations

- **VCUG (voiding cystourethrogram)**: size, VUR, emptying, **paradoxical filling** during voiding.
- **Cystoscopy**: enter and inspect the sac (flexible scope); cytology; biopsy carefully (perforation risk).
- **CT (computed tomography) urography**: tumour, stones, ureteric deviation, hydronephrosis.
- **Urodynamics**: BOO or neurogenic dysfunction (urine "sinks" into the sac, so contractility looks falsely low).

Management

- **Treat the cause first** (BOO); a small diverticulum may then empty and need only surveillance.
- **Observation** (cystoscopy, cytology) if asymptomatic; **CIC (clean intermittent catheterisation)** if unfit with poor emptying.
- **Endoscopic**: neck incision or resection ± fulguration in frail patients.
- **Diverticulectomy indications**: refractory symptoms, recurrent UTI, stones, upper-tract damage or VUR, premalignant change.

- **Tumour inside**: partial or radical cystectomy.

Diverticulectomy technique

- Open, laparoscopic or robotic; endoscopic prostate surgery first, same session.
- Steps: ureteric catheter if near the ureter → cystotomy → evert or dissect the sac from its pseudocapsule → close in two layers → catheter about 1 week, cystogram before removal.
- Complications: ureteric injury, leak.

Diagram to draw: paraureteral (Hutch) diverticulum with narrow neck and ureter.

> **Key points to remember:**
> - No muscle → stasis, UTI, stones, cancer.
> - Acquired type: treat the BOO first.
> - Tumour invading beyond lamina propria is T3a.

II-3d. Botulinim neurotoxin.

Type: SAQ (10 marks)

Assumed meaning: botulinum neurotoxin (spelling error in the question).

Introduction

- Toxin of *Clostridium botulinum*; serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**; type B is short-acting.
- 150 kDa: heavy chain (100 kDa) plus zinc-dependent light chain (50 kDa).

Mechanism

1. Heavy chain binds the cholinergic nerve terminal; the toxin is internalised.
2. Light chain enters the cytosol.
3. It **cleaves SNAP-25 (synaptosomal-associated protein 25)**, a SNARE (vesicle-docking) protein, so vesicles cannot fuse and acetylcholine release stops.

- **Afferent action**: less ATP (adenosine triphosphate) and substance P release; fewer TRPV1 and P2X3 receptors, so urgency falls too.
- Effect wears off as nerve terminals sprout (about 6–9 months).

Urological uses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U**; incontinence down about 59%, about 25% dry. ABC trial: equal to anticholinergic, more complete dryness. ROSETTA: 200 U equal to SNM (sacral neuromodulation), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY trials): lower detrusor pressure, larger capacity
Children with NDO	About 10 U/kg (maximum 360 U)
DSD (detrusor–sphincter dyssynergia)	Intrasphincteric injection in men unable to do CISC (clean intermittent self-catheterisation)
Others	Bladder pain syndrome; prostate injection for benign enlargement failed versus sham

Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor at **about 20 sites (100 U) or 30 sites (200 U)**, sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

Side effects

- **Retention needing CISC**: 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; higher in NDO. Each 10 mL rise in baseline PVR raises risk 27%. Patients must accept CISC beforehand.
- **UTI** about 20%; haematuria; rarely generalised weakness.
- Contraindications: active UTI, myasthenia gravis, unwillingness to self-catheterise.
- Repeat injections when the effect fades; response is maintained.

Diagram to draw: cholinergic synapse showing SNAP-25 cleavage.

Key points to remember:

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Main risks: retention needing CISC, UTI.

II-3e. Catheter free trial

Type: SAQ (10 marks)

Assumed meaning: trial without catheter (TWOC) after acute urinary retention.

Definition

- **TWOC (trial without catheter)**: catheter removal after decompression to test whether the patient voids adequately.
- **AUR (acute urinary retention)**: sudden, painful inability to void; commonest in older men with LUTS (lower urinary tract symptoms).

Types of AUR

Spontaneous	Precipitated
BPH (benign prostatic hyperplasia) progression	Anaesthesia, surgery, UTI (urinary tract infection), overdistension, anticholinergic or sympathomimetic drugs
TWOC fails more often; about 75% later need surgery	Better TWOC outcome; about a quarter need surgery

Initial management

- Catheterise; record drained volume; check creatinine; treat the trigger.
- Start an **α1-blocker** (alfuzosin or tamsulosin).

Rationale for α-blockers

- Sympathetic bladder-neck tone (dynamic component) adds to retention; α1-blockers relax it.
- **ALFAUR (Alfuzosin in Acute Urinary Retention)**: alfuzosin 10 mg for 3 days → TWOC success **62% vs 48%**.
- Meta-analysis: success relative risk about 1.5; fewer recurrences.

Procedure

- Remove the catheter after **2–3 days of α-blocker**, usually in the morning (no evidence favours short or long catheterisation).
- Encourage fluids; record voided volumes and **PVR (post-void residual)** by bladder scan.
- **Success**: comfortable voiding, low residual (unit threshold). **Failure**: recatheterise or start CISC (clean intermittent self-catheterisation).

Predictors of failure

- Advanced age; **drained volume >1 L**; spontaneous AUR; severe LUTS.
- Ultrasound: **IPP (intravesical prostatic protrusion) ≥10 mm**; thick bladder wall.

Results

- Success about **60%**; a second or third TWOC still succeeds in over a quarter.
- Recurrence after success is common, mostly within **6 months**; then offer surgery.

After TWOC

- **Success**: continue α-blocker; add a **5-ARI (5α-reductase inhibitor)** for large glands, because 5-ARIs, not α-blockers, prevent further AUR.
- **Failure**: TURP (transurethral resection of prostate) or other surgery; CISC or long-term catheter if unfit.
- **Also know:** immediate surgery has higher morbidity; TWOC also follows prostate surgery.

Diagram to draw: algorithm: AUR → catheter + α-blocker → TWOC at 2–3 days → success / failure.

Key points to remember:

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

II-3f. Urethral caruncle.

Type: SAQ (10 marks)

Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

Differential diagnosis

Lesion	Distinguishing feature
Urethral mucosal prolapse	**Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women
Urethral carcinoma	Firm, ulcerated, indurated, fixed; bleeding mass; biopsy
Urethral diverticulum	Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging)
Skene gland cyst or abscess	Lateral to meatus; distorts it; does not communicate
Others	Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus

- Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied.

Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions:

1. Grasp the tip and apply traction to expose the base.
2. Excise at the base.
3. Reapproximate mucosa with absorbable sutures if the defect is large.
4. Foley catheter for a few days.

- Complications: bleeding, recurrence, meatal stenosis.

Diagram to draw: meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

Rapid Revision Sheet – Paper II

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

II-1. CA bladder, etiology, newer investigations and treatment. (LAQ)

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

II-2. Describe etiology, clinical presentation and management of urethral strictures. (LAQ)

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Assess location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU only for bulbar <1.5 cm at first attempt; do not repeat.
- EPA is gold standard; BMG onlay for longer; staged repair for LS.

II-3a. Trans Uretero Ureteric anastomoses (SAQ)

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
- Risk: unilateral injury becomes bilateral.

II-3b. Acquired renal cystic disease (SAQ)

- ≥3 cysts per kidney in ESRD; >90% after 10 years.
- Uraemia causes it; transplant may regress cysts, not risk.
- RCC risk 3 times; image dialysis patients with haematuria.

II-3c. Bladder diverticulum. (SAQ)

- No muscle → stasis, UTI, stones, cancer.
- Acquired type: treat the BOO first.
- Tumour invading beyond lamina propria is T3a.

II-3d. Botulinim neurotoxin. (SAQ)

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Main risks: retention needing CISC, UTI.

II-3e. Catheter free trial (SAQ)

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

II-3f. Urethral caruncle. (SAQ)

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.